# Dongyoung Kim, Ph.D.

Artificial intelligence for science and engineering

AI professional @ Data Analytics Laboratory,
Samsung Life Insurance, Seoul, Republic of Korea
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## EXPERIENCE

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|---|---|
| 2020 – Current | **Samsung Life Insurance**, South Korea<br>AI professional at Data Analytics Laboratory. |
| 2017 – 2019 | **Institute for Basic Science**, South Korea<br>Research Fellow, Advisor: Prof. Yoon-Kyung Cho (Ulsan National Institute of Science and Technology). |
| 2012 – 2016 | **Texas A&M University,** U.S.A.<br>Research Assistant, Advisor: Raimund Ober (Texas A&M University). |
| 2010 – 2012 | **The University of Southwestern medical center**, U.S.A.<br>Research Assistant, |
| 2010 – 2012 | **The University of Texas at Dallas**, U.S.A.<br>Research Assistant |
| 2010 – 2010 | **EOS Australia,** Sydney, Australia<br>Intern |
| 2009 – 2010 | **Embedded software research center,** South Korea<br>Research Assistant |

## EDUCATION

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|---|---|
| 2012 – 2016 | **Ph.D., Biomedical Engineering,**<br>**Texas A&M University**<br>College station, Texas, USA |
| 2010 – 2012 | **B.S., Electrical engineering and computer science**<br>**The University of Texas at Dallas**<br>Richardson, Texas, USA |

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| 2006 – 2012 | **B.S. Electrical engineering and computer science**<br>**Kyungpook National University**<br>Daegu, South Korea |

## PROJECTS

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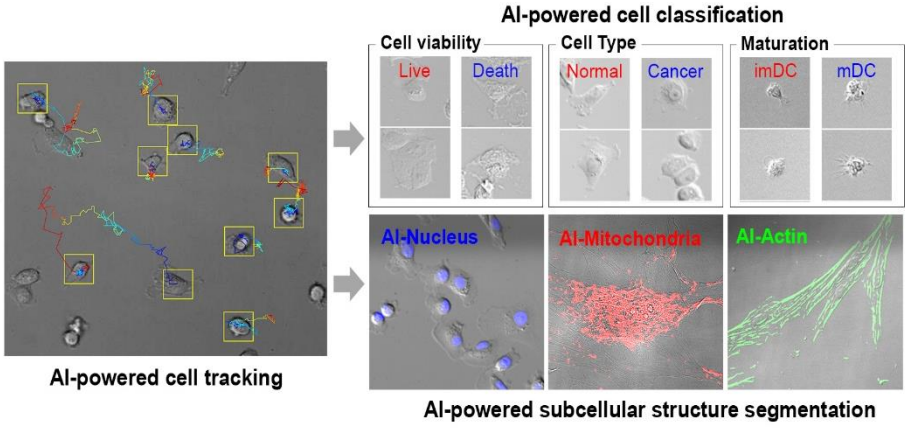
| 2020 – Currnet | **End-to-End deep learning powered optical character recognition framework**<br>- A set of deep learning models to understand document from document images.<br>- Models include generative adversarial networks, convolutional neural networks, recurrent neural networks and others for image conversions, corrections, recognition, context understanding and more.<br>- Two Korea patents are under preparation<br>- Two manuscripts are under preparation |
| 2020 – 2020 | **Deep learning powered data simulator for optical character recognition models.**<br>- Image simulator using generative adversarial network to train (or fake) deep learning models used for optical character recognition.<br><br>**Patent**<br>- **Kim D** et al. **Method for training data for optical character recognition.** Korea Patent Pending, 1020200059652, 2020 |
| 2020 – 2020 | **AI data/model management framework**<br>- A software framework to manage data and models of deep learning model training/inference.<br>- Optical data storage and transaction using light memory database.<br>- Model, experiment management and model production/serving using MLFlow service. |

2017– 2019 **AI-powered transmitted light microscopy for functional analysis of live cells.**




- Transforming a transmitted light microscopy image into a fluorescence microscopy equivalent image (or information) using unsupervised machine learning and supervised semantic segmentation.
- Real-time monitoring of living cell property and status from transmitted light microscopy video using region proposal convolutional neural network and convolutional neural network.
- Living cell tracking in time-lapse video using complementary learning of object shape and colors online.

Patents

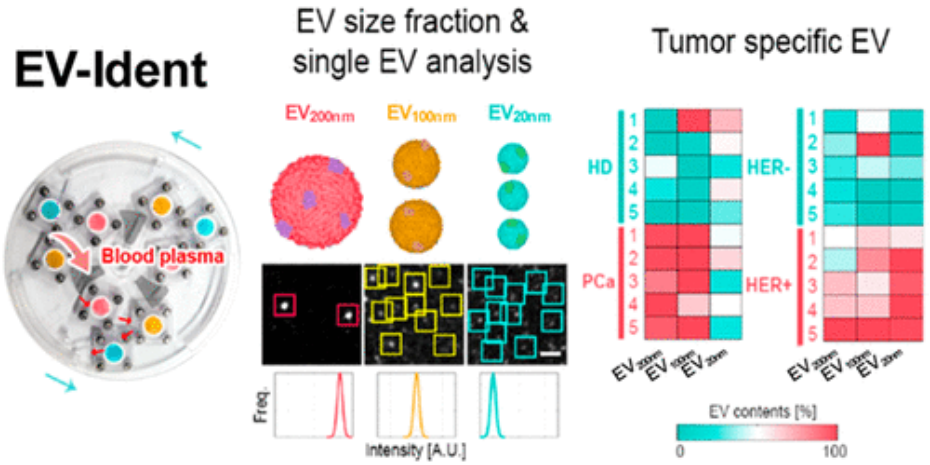
- **Kim D** et al. Generating method for specialized microscope images using artificial neural network and image processing apparatus. Korea Patent, 1020846820000, 2020.
- **Kim D** et al. Analysing method for cell image using artificial neural network and image processing apparatus for cell image. Korea Patent 1020846830000, 2020.

Publications

- **Kim, D.**, Min, Y., Oh, J.M. et al. **AI-powered transmitted light microscopy for functional analysis of live cells**. Sci Rep 9, 18428 (2019) doi:10.1038/s41598-019-54961-x

2017 – 2019

EV-Ident: Identifying Tumor-Specific Extracellular Vesicles by Size Fractionation and Single-Vesicle Analysis



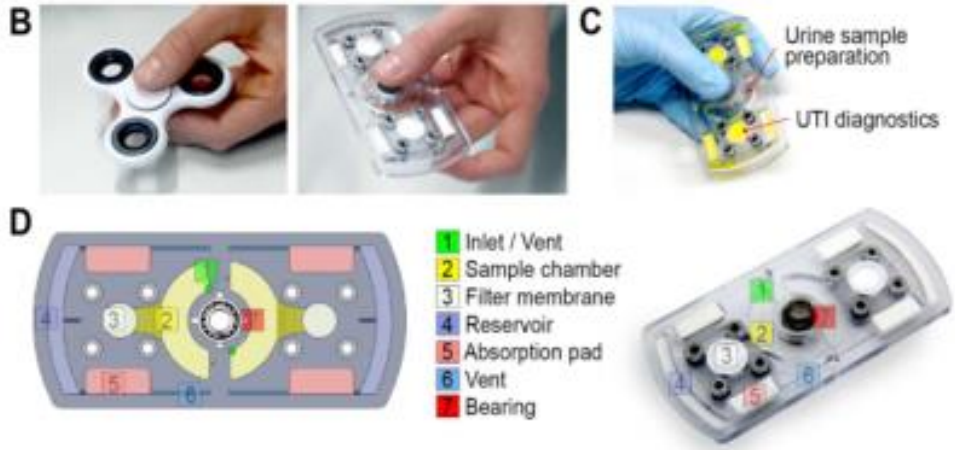

- Early cancer prediction from blood born particle using machine learning techniques.

Publications

- **Kim, D.,** Woo, H.-K., Lee, C., Min, Y., Kumar, S., Sunkara, V., … Cho, Y.-K. (2020). **EV-Ident: Identifying Tumor-Specific Extracellular Vesicles by Size Fractionation and Single-Vesicle Analysis.** Analytical Chemistry, 92(8), 6010–6018. https://doi.org/10.1021/acs.analchem.0c00285

2017 – 2019

A fidget spinner for the point-of-care diagnosis of urinary tract infection




- Hand-powered microfluidic system for bacterial cell enrichment.
- Urinary tract infection diagnosis from a cell phone photograph of a testing kit with machine learning techniques.

Patents

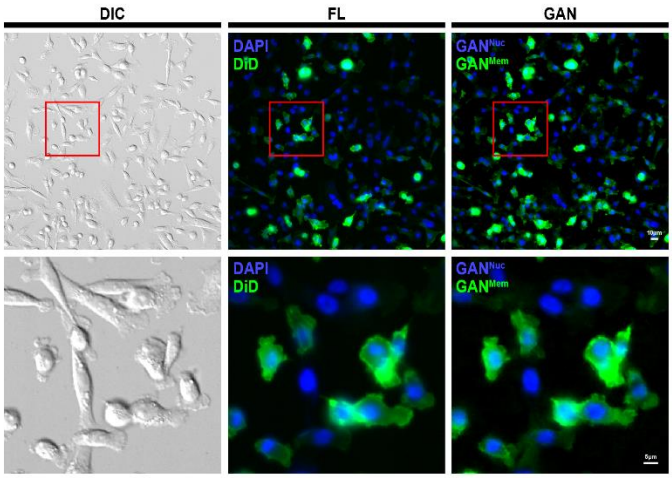
- **Kim D** et al. Centrifugal force based non-powered particle concentration apparatus and method of particle concentration. Korea Patent 1021037840000, 2020.
- **Kim D** et al. Centrifugal force based platelet isolation and testing system. Korea Patent 1020638650000, 2020.

Publications

- **Kim, D.,** Michael, I., Gulenko, O., Kumar, S., Kumar, S., Clara, J., … Cho, Y.-K. (2020). **A fidget spinner for the point-of-care diagnosis of urinary tract infection.** Nature Biomedical Engineering. https://doi.org/10.1038/s41551-020-0557-2

2019 – Current

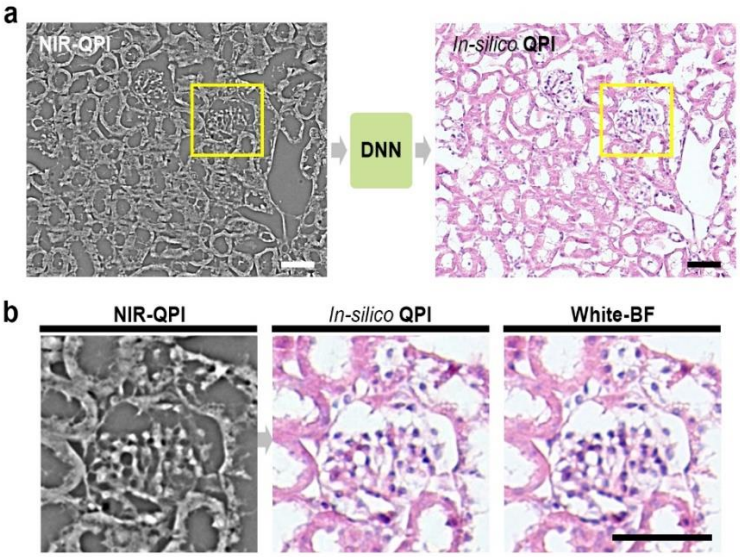
**In-silico fluorescence staining using generative adversarial network.**




- A deep learning-based approach to create fluorescence stained image from the optical diffraction patterns of the sample.
- Manuscript under preparation.

2019 – Current

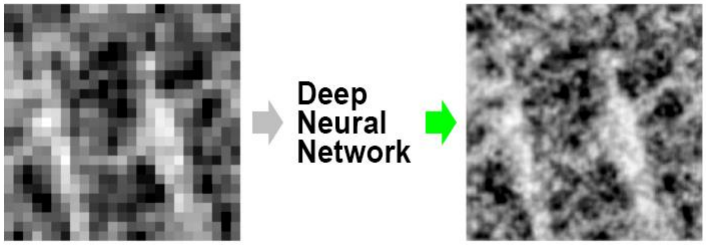
**In-silico histology staining using generative adversarial network.**




- A deep learning-based approach to create histology stained image from the optical phase information of the unstained slide sample.
- Manuscript under preparation.

2019 – Current

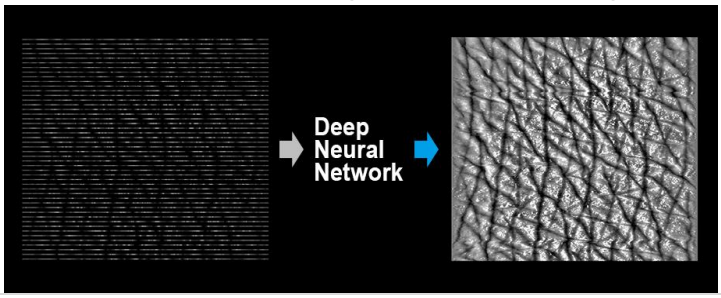
**Generative adversarial network for high definition medical image recovery.**




- A deep learning-based approach to create high-resolution tomography images acquired from a low-powered device.
- Manuscript under preparation.

2019 – Current

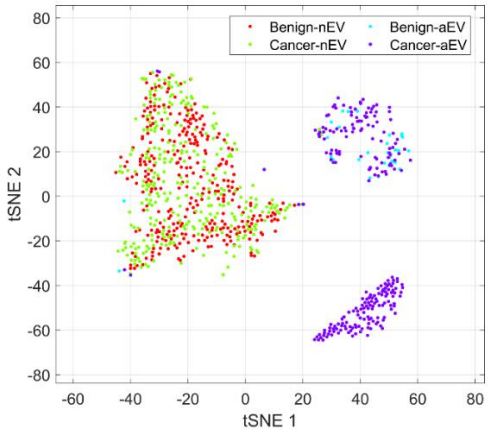
**Generative adversarial network for rapid medical data acquisition.**



- A deep learning-based approach to make a medical tomography imaging 10 times faster.
- Manuscript under preparation.

2019 – Current

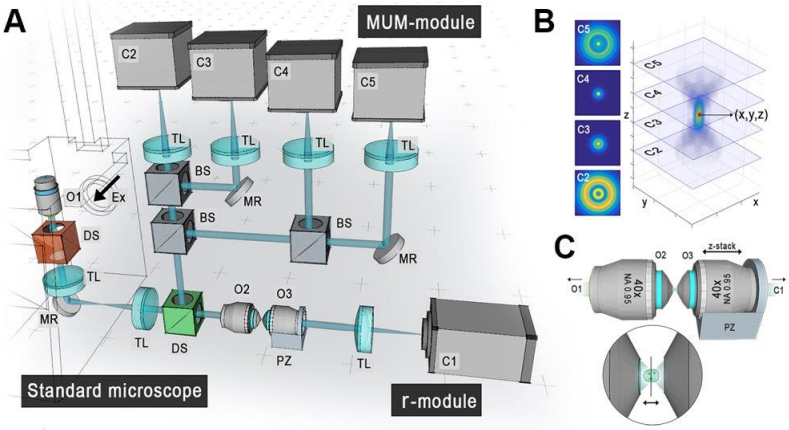
**Searching extreme rare cancer extracellular vesicle in blood plasma using generative adversarial network.**




- A deep learning-based approach to detect ultra-rare cancer signal from blood.
- Manuscript under preparation.

2014 – 2016

**Remote focusing multifocal plane microscopy for the imaging of 3D single molecule dynamics with cellular context**




- A microscopy modality to visualize three-dimensional single molecule trajectory in a context of three-dimensional and multicolor cellular structures and compartments over time

- Estimating three-dimensional single molecule location in ~10 nm accuracy.
- Imaging three-dimensional cellular structures from an optical replica using relay optics.

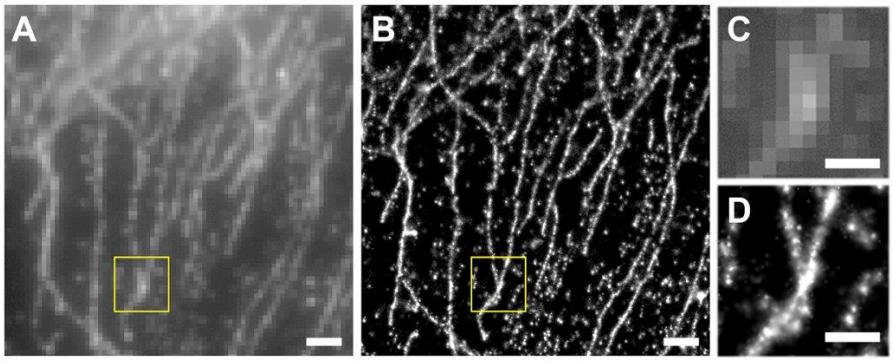
Patents

- **Kim D.,** Ober R.J, **Advanced multi-dimensional microscopy system for single particle & structure imaging.** U.S. Patent, 2017.)

Publications

- **Kim, D.,** Chao, J., Velmurugan, R., You, S., Ward, E. S., and Ober, R. J. **Remote focusing multifocal plane microscopy for the imaging of 3D single molecule dynamics with cellular context.** Proceedings of the SPIE, Three-Dimensional and Multidimensional Microscopy: Image Acquisition and Processing XXIV, 10070: 2017.
- **Kim D**, You S, Ward E.S., Ober R.J, **Imaging of three-dimensional single molecule dynamics with cellular context: Antibody trafficking and interaction with cell membrane and sorting endosomes.** ASCB, San Fransisco, CA. 2016.

2012 – 2015 **Three-dimensional super resolution microscopy**




- An optical microscopy technique resolving fluorescently labeled features in ~5 nm resolution.
- Multifocal plane microscopy enables three-dimensional super-resolving imaging.

- Multicolor imaging and super-precision chromatic/spatial registration are capable.

Publications

- Cohen E. A. K.,. **Kim D.**, Ober R.J, **Cramer-Rao Lower Bound for Point Based Image Registration with Heteroscedastic Error Model for Application in Single Molecule Microscopy.** IEEE Transactions on Medical Imaging 2015
- Ram S, **Kim D**, Ober RJ, Ward ES. **3D Single Molecule Tracking with Multifocal Plane Microscopy Reveals Rapid Intercellular Transferrin Transport at Epithelial Cell Barriers**. Biophysical Journal 2012

## PATENTS

**Kim D** et al. **Method for training data for optical character recognition.** Korea Patent Pending, 1020200059652, 2020

**Kim D** et al. **Generating method for specialized microscope images using artificial neural network and image processing apparatus**. Korea Patent, 1020846820000, 2020.

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**Kim D** et al. **Centrifugal force based non-powered particle concentration apparatus and method of particle concentration.** Korea Patent 1021037840000, 2020.

**Kim D** et al. **Centrifugal force based platelet isolation and testing system.** Korea Patent 1020638650000, 2020.

**Kim D.,** Ober R.J, **Advanced multi-dimensional microscopy system for single particle & structure imaging.** U.S. Patent, 2017.

## HONORS & AWARDS

| | |
|---|---|
| 2018 | 1st place at Shark Tank Competition (Startup company competition).<br>**The 22nd International Conference on Miniaturized Systems For Chemistry And Life Sciences** |
| 2016 | 1st place at 3D single molecule localization microscopy<br>**SMLMS Challenge, EPFL, Switzerland.**<br>*Software developed using **MATLAB** |

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|---|---|
| 2014 – 2016 | Scholarship, **Texas A&M University, U.S.A.** |
| 2012 – 2014 | Scholarship, **The University of Texas at Dallas, U.S.A.** |
| 2011 – 2011 | Scholarship, **The University of Texas at Dallas , U.S.A.** |
| 2011 – 2011 | 2nd place for Senior Design Project, **The University of Texas at Dallas** |
| 2010 – 2012 | Scholarship, **The University of Texas at Dallas, U.S.A.** |

## LANGUAGES

**English, Korean**

## SKILLS

| | |
|---|---|
| **Artificial Intelligence** | Supervised/unsupervised machine learning, Deep learning model development, Data & modal management framework and platform design. |
| **Programming** | MATLAB, Python, C, C++, JAVA. |
| **Data science** | Big data analysis, Optimization based data processing. |
| **Electrical system architecture** | NI LabWindows/CVI, NI LabView. |
| **Optical system architecture** | Zemax based optical system design, Multimodal optical system configuration. |
| **Microscope** | Epifluorescence, TIRF microscopy, Confocal microscopy<br>Super-resolution microscopy<br>Single molecule microscopy<br>Transmitted/scanning electron microscopy |

## LECTURES

**BME21101 Introduction to biomedical engineering** (**Introduction to artificial intelligence for biomedical engineering / Introduction to deep learning**), 2019, Ulsan National Institute of Science and Technology (UNIST).

## PUBLICATIONS

**Kim, D.,** Min, Y., Oh, J.M. et al. **AI-powered transmitted light microscopy for functional analysis of live cells.** Sci Rep 9, 18428, 2019

**Kim, D.,** Woo, H.-K., Lee, C., Min, Y., Kumar, S., Sunkara, V., … Cho, Y.-K. (2020). **EV-Ident: Identifying Tumor-Specific Extracellular Vesicles by Size Fractionation and Single-Vesicle Analysis.** Analytical Chemistry, 92(8), 6010–6018.

**Kim, D.,** Michael, I., Gulenko, O., Kumar, S., Kumar, S., Clara, J., … Cho, Y.-K. (2020). **A fidget spinner for the point-of-care diagnosis of urinary tract infection.** Nature Biomedical Engineering.

Sunkara, V., Kim C., Park J., Woo H., **Kim D.**, Ha H., Kim M., Son Y., Kim J., Cho Y., **Fully automated, label-free isolation of extracellular vesicles from whole blood for cancer diagnosis and monitoring.** Theranostics, 2019.

Michael I., Kumar S., Oh J,, **Kim D**, Kim J., Cho Y. **Surface Engineered Paper Hanging Drop Chip for 3D Spheroid Culture and Analysis.** ACS applied materials & interfaces, 2018.

Li, R., Chiguru, S., Li, L., **Kim, D.**, Velmurugan, R., Kim, D., Tian, H., Schroit, A., Mason, R., Ober, R. J. and Ward, E. S. **Targeting phosphatidylserine with calcium-dependent protein-drug conjugates for the treatment of cancer.** Molecular Cancer Therapeutics, 2018.

Devanaboyina, S. C., Khare, P., Challa, **Kim, D.,** Ober, R. J., and Ward, E. S. **Engineered clearing agents for the selective depletion of antigen-specific antibodies.** Nature Communications, 2017.

**Kim, D.,** Min, Y., Oh, J.M. et al. **AI-powered transmitted light microscopy for functional analysis of live cells.** Sci Rep 9, 18428, 2019

**Kim, D.,** Woo, H.-K., Lee, C., Min, Y., Kumar, S., Sunkara, V., … Cho, Y.-K. (2020). **EV-Ident: Identifying Tumor-Specific Extracellular Vesicles by Size Fractionation and Single-Vesicle Analysis.** Analytical Chemistry, 92(8), 6010–6018.

**Kim, D.,** Michael, I., Gulenko, O., Kumar, S., Kumar, S., Clara, J., … Cho, Y.-K. (2020). **A fidget spinner for the point-of-care diagnosis of urinary tract infection.** Nature Biomedical Engineering.

Sunkara, V., Kim C., Park J., Woo H., **Kim D.**, Ha H., Kim M., Son Y., Kim J., Cho Y., **Fully automated, label-free isolation of extracellular vesicles from whole blood for cancer diagnosis and monitoring.** Theranostics, 2019.

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Devanaboyina, S. C., Khare, P., Challa, **Kim, D.,** Ober, R. J., and Ward, E. S. **Engineered clearing agents for the selective depletion of antigen-specific antibodies.** Nature Communications, 2017.

**Kim, D.,** Chao, J., Velmurugan, R., You, S., Ward, E. S., and Ober, R. J. **Remote focusing multifocal plane microscopy for the imaging of 3D single molecule dynamics with cellular context.** Proceedings of the SPIE, Three-Dimensional and Multidimensional Microscopy: Image Acquisition and Processing XXIV, 10070: 2017.

Vahid, M. R., Chao, J., **Kim, D.,** Ward, E. S., and Ober, R. J. **State space approach to single molecule localization in fluorescence microscopy.** Biomedical Optics Express, 2017

Poovasery J.S., Kang J.C., **Kim D**, Ober R.J, Ward E.S. **Antibody targeting of HER2/HER3 signaling overcomes heregulin-induced resistance to PI3K inhibition in prostate cancer.** International Journal of Cancer 2017

**Kim D**, You S, Ward E.S., Ober R.J, **Imaging of three-dimensional single molecule dynamics with cellular context: Antibody trafficking and interaction with cell membrane and sorting endosomes.** ASCB, San Fransisco, CA. 2016.

Cohen E. A. K.,. **Kim D**., Ober R.J, **Cramer-Rao Lower Bound for Point Based Image Registration with Heteroscedastic Error Model for Application in Single Molecule Microscopy.** IEEE Transactions on Medical Imaging 2015

Ram S, **Kim D**, Ober RJ, Ward ES. **The level of HER2 expression is a predictor of antibody-HER2 trafficking behavior in cancer cells**. mAbs 2014

Devanaboyina SC, Lynch SM, Ober RJ, Ram S, **Kim D**, Puig-Canto A, Breen S, Kasturirangan S, Fowler S, Peng L, et al. **The effect of pH dependence of antibody-antigen interactions on subcellular trafficking dynamics**. mAbs 2013

Ram S, **Kim D**, Ober RJ, Ward ES. **Microscopy in Polarized Epithelia Reveals a Novel Cellular Process of Intercellular Transfer.** Biophysical Journal 2013

Ram S, **Kim D**, Ober RJ, Ward ES. **3D Single Molecule Tracking with Multifocal Plane Microscopy Reveals Rapid Intercellular Transferrin Transport at Epithelial Cell Barriers**. Biophysical Journal 2012

**Kim D**, Ram S, Ober RJ, Ward ES. **3D single molecule tracking of rapid intracellular trafficking imaged by multifocal plane microscopy** Microscopy and Microanalysis 2012.